SOFTWARE VALIDATION & VERSION HISTORY REPORT

PROJECT : *CalcSuite Pro: Digi-ToolKit*

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STATUS : *Final Release (v7.6)*

1. PROJECT OVERVIEW

CalcSuite Pro is a multi-tool mathematical workstation featuring secure Abstract Syntax Tree (AST) evaluation, Cramer's Rule matrices, and a custom polynomial dictionary engine for natural-language algebra parsing.

2. MASTER TEST CASES & VALIDATION SUITE

The following cases were executed to validate the engine's logical accuracy and error handling.

TAB 1: SCIENTIFIC CALCULATOR

- **Standard Arithmetic:** `2 * (5 + 5)`
  - **Expected Result:** `20`
- **Implicit Multiplier:** `2pi`
  - **Expected Result:** `6.2831853`
- **Trigonometry (DEG):** `sin(90)`
  - **Expected Result:** `1.0`
- **Advanced Logic:** `FIBS(10)`
  - **Expected Result:** `Popup window displaying 0, 1, 1, 2, 3, 5, 8, 13, 21, 34`
- **Syntax Guard:** `2x + 5`
  - **Expected Result:** `450ms human-speed delay -> Redirects to Algebra Mode.`

TAB 2: SOLVERS

- **Quadratic (Real):** `a=1, b=5, c=6`
  - **Expected Result:** `x1 = -2, x2 = -3`
- **Quadratic (Complex):** `a=1, b=2, c=5`
  - **Expected Result:** `x1 = -1 + 2i, x2 = -1 - 2i`
- **Linear Fallback:** `a=0, b=2, c=-10`
  - **Expected Result:** `Triggers 'Linear Fallback' note -> x = 5`
- **Linear 2x2 (Unique):** `EQ I: 2,3,10 | EQ II: 4,-1,6`
  - **Expected Result:** `x = 2, y = 2`
- **Linear (No Solution):** `EQ I: 2,4,10 | EQ II: 2,4,20`
  - **Expected Result:** `Triggers 'Parallel Lines / Inconsistent' system guard.`

TAB 3: SMART ALGEBRA

- **Numeric Bypass:** `5 + 10 * 2`
  - **Expected Result:** `Result: 25`
- **Simplification:** `2x + 5x - x`
  - **Expected Result:** `Simplified: 6x`
- **Linear (1 Var):** `3x - 5 = 10`
  - **Expected Result:** `Rearranged: 3x - 15 = 0 -> x = 5`
- **Linear (2 Var):** `2a - 6b = 19, 4a + 4b = 12`
  - **Expected Result:** `a = 4.625, b = -1.625`
- **Bracket Expansion:** `(x-2)(x+5)(x-1)(x+1) = 6`
  - **Expected Result:** `Polynomial iteration -> Rearranged -> +4 = 0`
- **Polynomial Guard:** `x^3 + 2x = 0`
  - **Expected Result:** `Triggers Degree-3 analytical limitation note.`
- **Cross-Term Guard:** `2xy + 5 = 0`
  - **Expected Result:** `Triggers multi-variable cross-term limitation note.`
- **Underdetermined System:** `7a+8b-9a+12b = 12a-13b+5a-8`
  - **Expected Result:** `Engine rearranges to -19a+33b+8 = 0 -> Triggers 'Underdetermined' guard -> Instructs user to provide a second equation.`
- **Format Guard (Non-Linear):** `(x-3)(x+4) = 2x`
  - **Expected Result: Triggers 'Format Error' ->** `Prompts user to move all variables to the left side to enforce standard polynomial form.`
- **Format Guard (Quadratic):** `x^2 + 4x = 2x + 7`
  - **Expected Result: Triggers 'Format Error' ->** `Prompts user to move all variables to the left side.`

TAB 4: HEALTH / BMI

- **Metric System:** Age 25, Male, 100kg, 180cm
  - **Expected Result:** BMI: 30.9 (Obese) + Athlete context note.
- **Imperial System:** Age 25, Female, 140lb, 5ft 5in
  - **Expected Result:** BMI: 23.3 (Normal) + Female body-fat context note.
- **Age Guard:** Age 15, Male, 60kg, 160cm
  - **Expected Result:** Calculates BMI but overrides category -> requires chart.

TAB 5: UNIT CONVERTER

- **Standard Conversion:** Length -> 1 Metre to Centimetre
  - **Expected Result:** 100 cm
- **Advanced Conversion:** Temperature -> 0 Celsius to Fahrenheit
  - **Expected Result:** 32 °F
- **Swap Functionality:** Press 'Swap' on previous result
  - **Expected Result:** Swaps to 32 Fahrenheit -> 0 Celsius.

3. EXTENDED VERSION HISTORY

This application underwent continuous daily iteration to refine security, logic handling, and user interface responsiveness.

Version	Date	Status	Key Upgrades & Architectural Changes
v1.0	30 March 2026	Legacy	Initial deployment. Basic arithmetic GUI logic established using standard libraries.
v1.2	01 April 2026	Legacy	Incremental logic optimizations and operational bug fixes.
v2.0	02 April 2026	Legacy	Major UI restructuring and visual layout adjustments.
v3.2	03 April 2026	Legacy	Multi-tool expansion. BMI and Unit Converter algorithms integrated.
v3.3	04 April 2026	Legacy	Refinement of converter parameters and unit scaling.
v4.0	07 April 2026	Legacy	Implementation of history tracking and state persistence.
v4.2	08 April 2026	Legacy	History UI optimizations and text formatting updates.
v5.0	09 April 2026	Legacy	Security & UI Overhaul. CustomTkinter implemented. Unsafe eval() replaced by strict AST parsing.
v6.0	10 April 2026	Legacy	Core solver engine mathematical foundation established.

v6.1	11 April 2026	Legacy	Quadratic solver edge-cases and imaginary number handling addressed.
v6.2	12 April 2026	Legacy	Linear 2x2 solver logic and Cramer's Rule matrix implementation.
v6.3	13 April 2026	Legacy	Error handling and validation deployment for the Solvers tab.
v7.0	15 April 2026	Stable	Smart Algebra logic introduced. Natural language parsing for single-variable equations.
v7.2	16 April 2026	Stable	Parsing improvements for algebraic terms and string extraction.
v7.3	20 April 2026	Stable	Simplification routing and numeric bypass logic integrated.
v7.4	21 April 2026	Stable	Iterative FOIL logic added for bracket expansion.
v7.5	22 April 2026	Stable	The "Pro" Overhaul. Digi-ToolKit branding applied. Smart cross-tab redirects built.
v7.6	24 April 2026	Final	The Stability Update. Dictionary-based polynomial engine implemented. Epsilon floating-point tolerance deployed. Legacy code fully pruned.